

C-346 — C Programming Fundamentals & Data Structures

OFFICIAL COURSE SYLLABUS & CURRICULUM GUIDE

Course Code: C-346	Category: Computer Science & Tech
Academic Term: 2026 Academic Year	Format: E-Learning / Self-Paced
Prerequisites: Basic Problem Solving	Estimated Hours: 40 Hours Content

1. Course Description & Learning Objectives

Welcome to **C Programming Fundamentals & Data Structures (C-346)**! This course is designed to equip students with deep theoretical understanding and practical hands-on proficiency. By completing this curriculum, learners will master key concepts, build industry-standard projects, and develop problem-solving skills necessary for professional software development.

Key Learning Outcomes:

- Master foundational and advanced topics through structured video lectures and interactive materials.
- Apply theoretical principles to real-world coding projects and algorithmic problem solving.
- Understand best practices, debugging strategies, and architectural design patterns.
- Pass comprehensive evaluations and earn official course certification.

2. Course Modules & Detailed Schedule

Module / Topic	Key Topics & Assignments	Weightage
Module 1: Introduction to C & GCC Compiler	In-depth lectures, reading materials, and hands-on coding exercises.	25%
Module 2: Variables, Primitive Types & Input/Output	In-depth lectures, reading materials, and hands-on coding exercises.	30%
Module 3: Control Flow & Iterative Statements	In-depth lectures, reading materials, and hands-on coding exercises.	35%
Module 4: Functions & Call by Value vs Reference	In-depth lectures, reading materials, and hands-on coding exercises.	40%
Module 5: Arrays, Strings & Character Buffers	In-depth lectures, reading materials, and hands-on coding exercises.	45%
Module 6: Pointers & Dynamic Memory Allocation	In-depth lectures, reading materials, and hands-on coding exercises.	50%

3. Grading Policy & Assessment Breakdown

Component	Percentage	Details
Practical Exercises & Quizzes	30%	Weekly online quizzes & code exercises
Mid-Term Assessment	30%	Comprehensive theoretical & coding test
Final Capstone Project	40%	Individual end-to-end practical project